

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458627.7306 +/- 0.0023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.189 +/- 0.017
Transit depth (Rp/Rs)^2	3.59 +/- 0.64 [%]
Orbital Inclination (inc)	85.41 +/- 0.19
Airmass coefficient 1 (a1)	0.9368 +/- 0.0094
Airmass coefficient 2 (a2)	0.049 +/- 0.018
Scatter in the residuals of the lightcurve fit is	1.01 %
Best Comparison Star	#1 - [470, 141]
Optimal Aperture	3.62
Optimal Annulus	11.46
Transit Duration (day)	0.0866 +/- 0.0061

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.189 $\pm$ 0.017 vs 0.1457 (deviation 2.04 σ)
Duration fit vs published	0.0866 d vs 0.12 d (deviation 4.4 σ)
Mid-time O-C	-23.0 min
Depth SNR	8.9
Residual scatter	1.01 %

Light curve

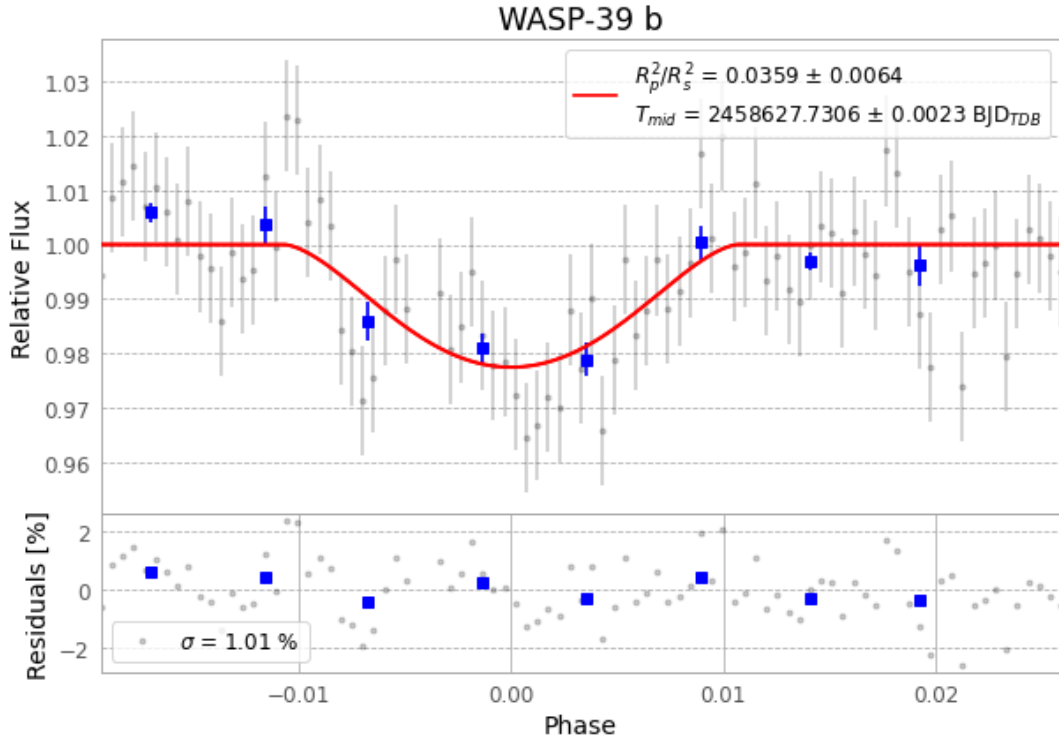


Figure 1 — FinalLightCurve_WASP-39 b_2019-05-23.png (SHA-256 0c9c8aa892754c77...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [326, 123], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 c7f52195c195a105...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

89 FITS frames (58 MB), WASP-39190524033012.FITS → WASP-39190524075412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-39-190524.log (dcff3ee20162...), FinalLightCurve_WASP-39 b_2019-05-23.png (0c9c8aa89275...), FinalLightCurve_WASP-39 b_2019-05-23.csv (61a674a4fe4d...)

Compute resources

Wall time 125.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 11:43Z.